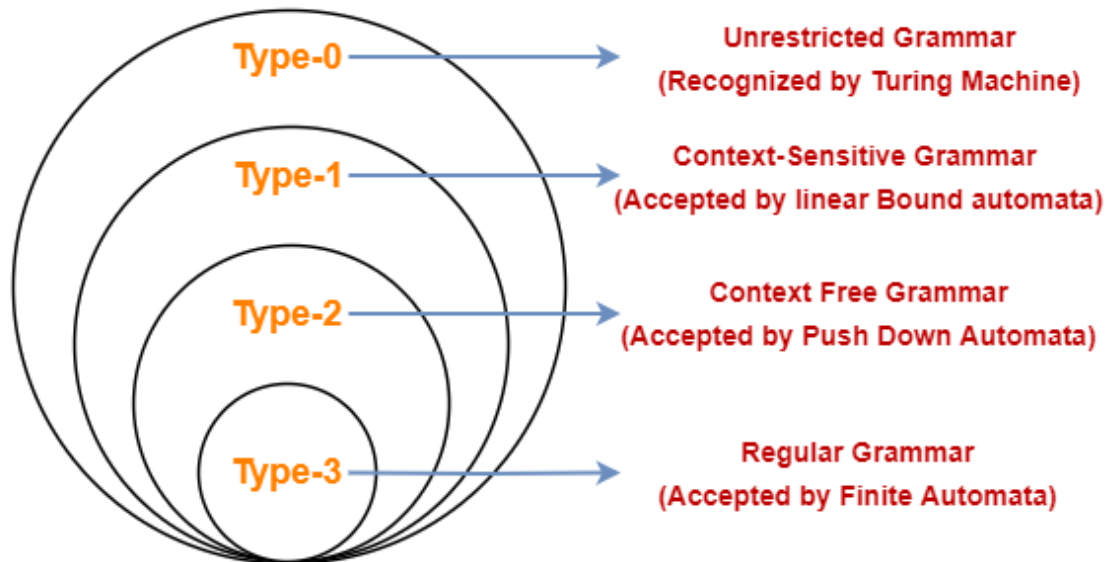


Grammars



Type 2	Context free grammars	PDA (Push down automaton), Non deterministic push down automaton	$A \rightarrow \alpha$	$L = \{a^n b^n \mid n > 0\}$
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Pushdown Automata

PDA M is defined as

$$M = (Q, \Sigma, \Gamma, \delta, s(=q_0), Z, F)$$

$Q = \{q_1, q_2, \dots, q_n\}$ **finite** set of states

$\Sigma = \{a, b, c, d, \dots, z\}$ **finite** set of input alphabets, can be any characters

$\Gamma = \{a, b, c, d, \dots, z\}$ **finite** set of **stack** alphabets, can be any characters

$\delta = Q \times (\Sigma \cup \{\epsilon\}) \times \Gamma \times Q \times \Gamma^*$ **finite** set of state transitions

$s(q_0) \in Q$ initial state

$Z \in \Gamma$ initial stack symbol

$F \subset Q$ (non-empty) set of final states

Properties of state transitions

$$\delta(p, a, A, q, \alpha) \in \delta$$

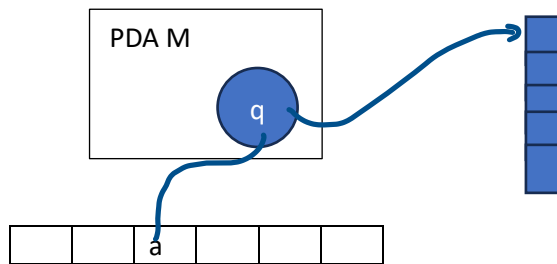
p is the state of the automaton

a is the next input alphabet

A is the top symbol of the stack

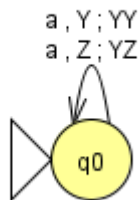
q is the state to transition to

α is the symbol that replaces symbol A on the stack (Push, Pop, Nop)



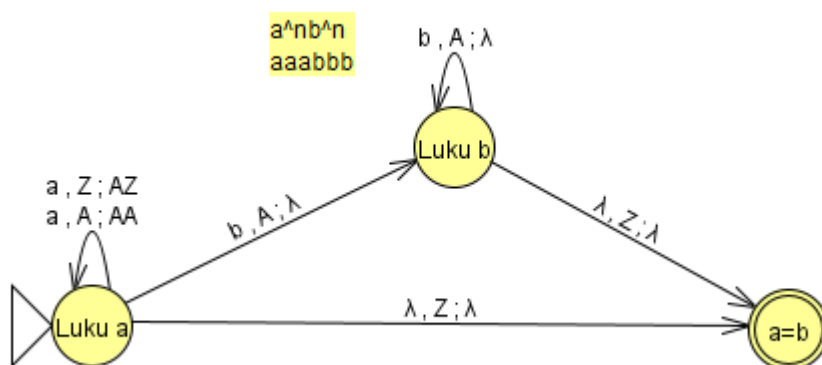
Push, Pop, Nop

- Always performs a Pop operation
- Nop is Push back
- Pop is λ
- Push is Push+Push



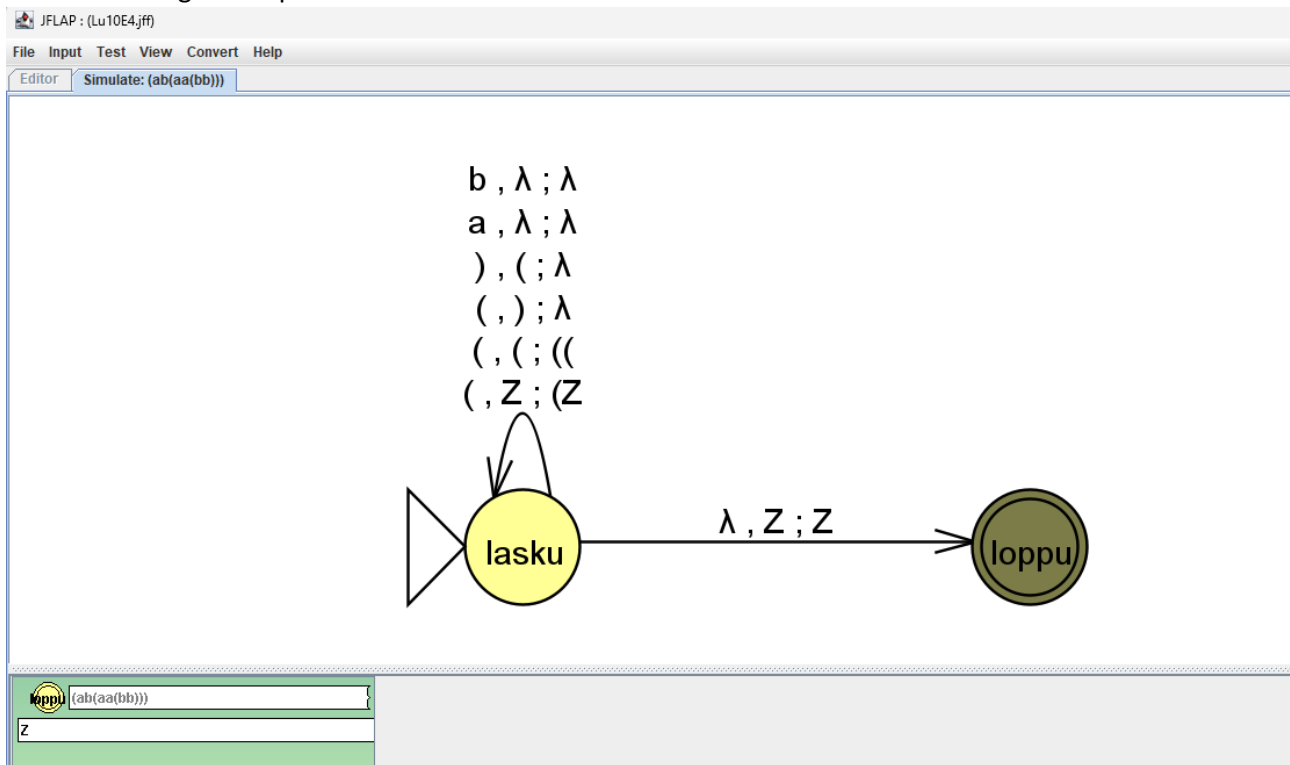
$a^n b^n$ PDA

Accepts input where there are as many a letters as b letters and they are arranged so that the a 's come first, for example $aaaabbbb$ is accepted but $abba$ is not



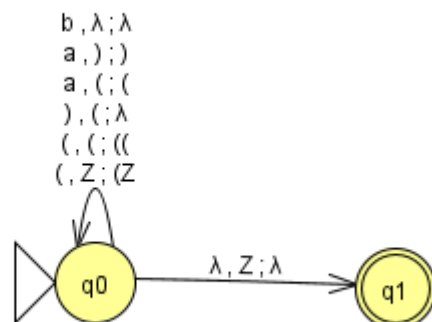
Programming and parsing

Version starting with a parenthesis:



Starting with a parenthesis or the letter a

(abbba(bb(aa))))



Palindrome (Nondeterministic)

